IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

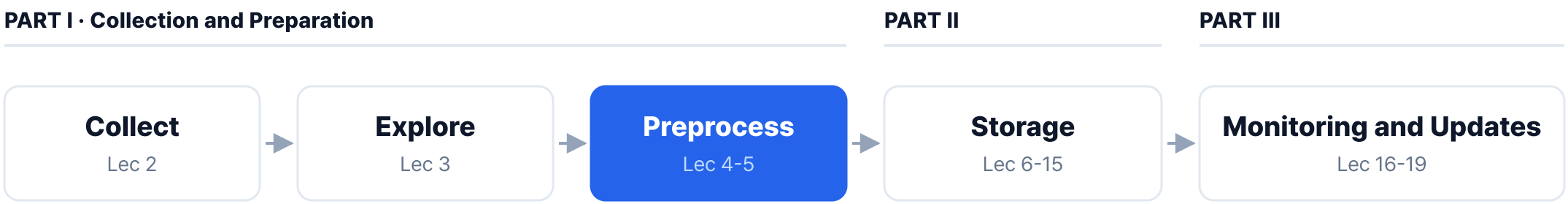
5.

Pre-processing II: Transformation & Reduction

Sangwoo Mo · POSTECH

Industrial & Management Engineering

Where We Are

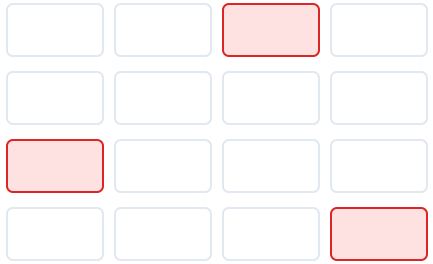


preprocessing changes the representation; some choices also remove information

Recap: Four Stages of Preprocessing

01

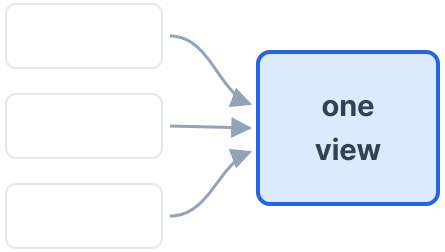
Cleaning



Wrong values in one table

02

Integration



Tables that disagree

03

Transformation



Values a model can read

04

Reduction



Less data, same signal

Same four stages as lecture 4. **Two done, two to go**

Transformation

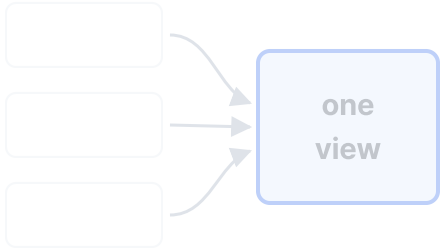
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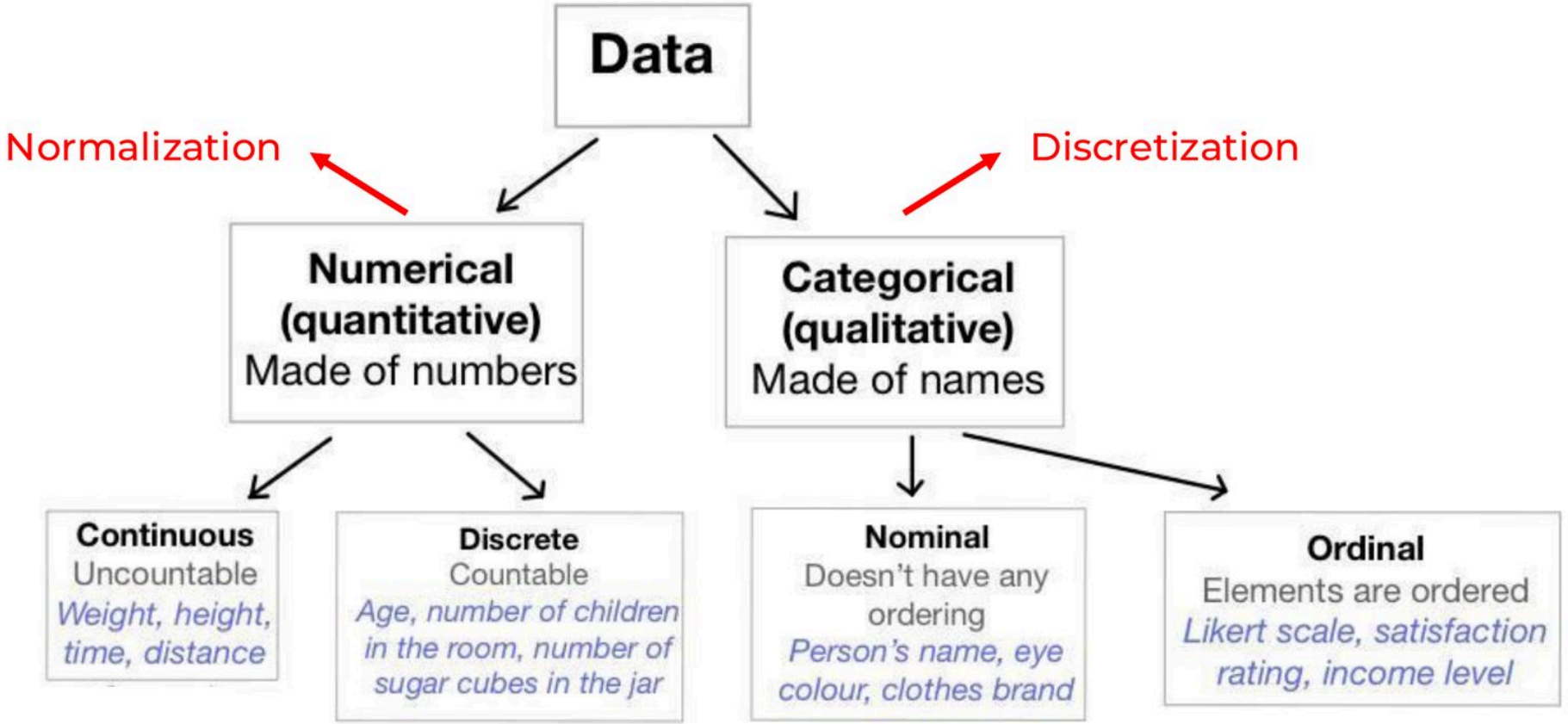
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Reduction

Less data, same signal

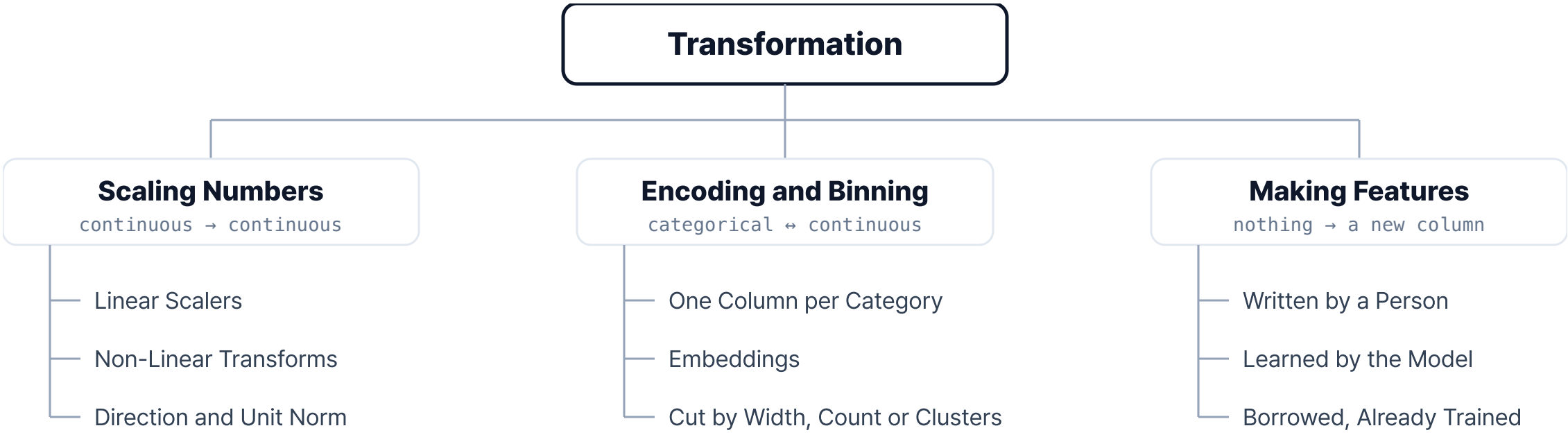
Recap: Data Types

Numbers get rescaled; discretization turns numbers into categories



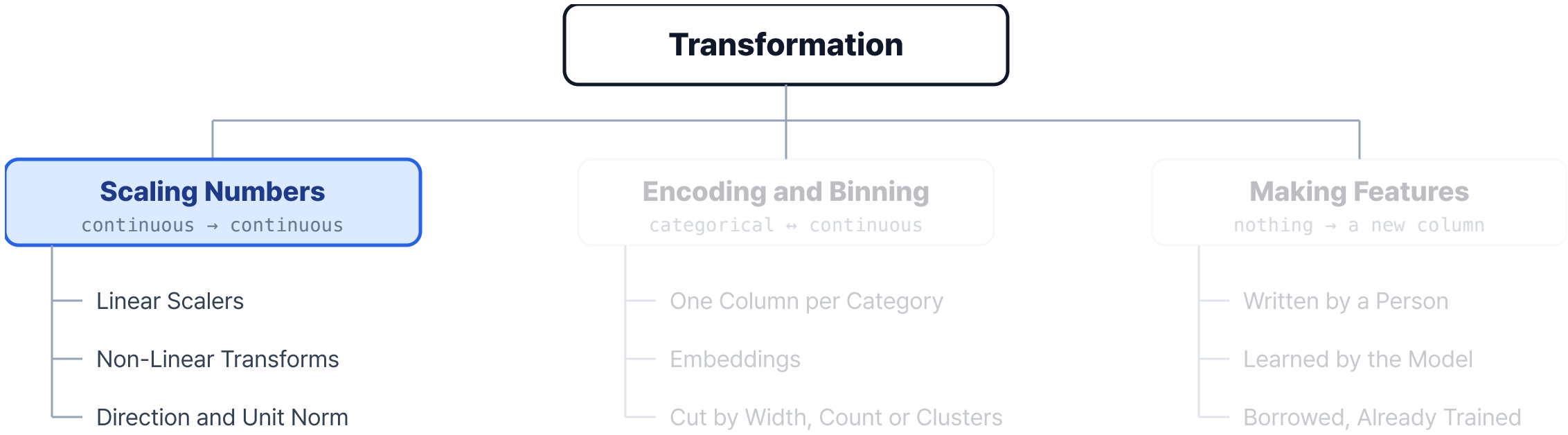
The type of a column **decides the tool**

Three Branches of Transformation



Scaling keeps the type; **encoding and binning cross it**; making features adds columns

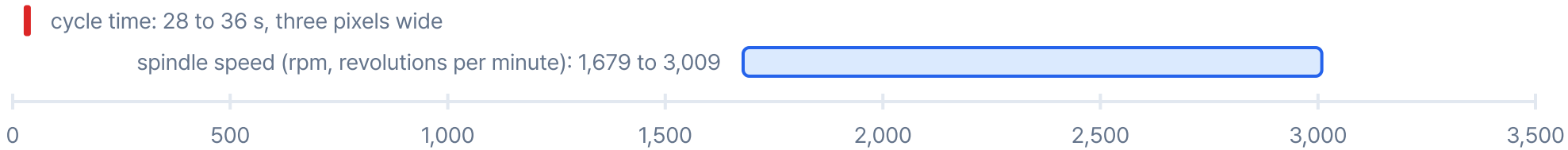
Scaling Numbers



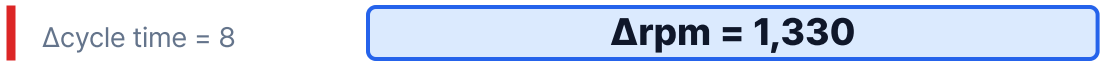
Why Scale at All

Distance-based models add up raw units, so the widest column decides

two columns from one machine, on one raw axis



how far apart two rows can be, column by column



In raw Euclidean distance, the rpm difference **dominates** the cycle-time difference

Fitted Transformation State

Fit learns parameters; transform reuses the saved parameters

Step	Fit on train	Transform new rows
Scaling	min, max or mean, SD	reuse the same numbers
Encoding	category vocabulary	reuse the same columns
PCA	center and axes	project onto the same axes

Fit once on training rows, then apply the saved state

Linear Scalers

Three scalers, one shape: subtract a center, divide by a spread

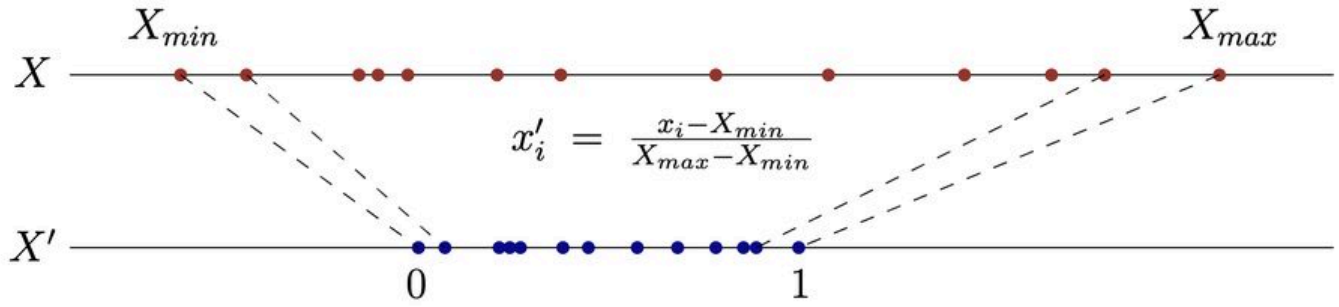
SCALER	THE SAME FORMULA, TWO ESTIMATORS	BREAKDOWN POINT
Min-max	(x − **min**) / **max − min**	0%
Standardization	(x − **mean**) / **standard deviation**	0%
Robust	(x − **median**) / **interquartile range**	25%

Picking a scaler is picking how many bad rows it takes to move the answer

Min-Max Scaling

x′ = (x − min) / (max − min) maps the training range onto [0, 1]

$$x_i' = \frac{x_i - X_{min}}{X_{max} - X_{min}}$$



The minimum maps to 0, the maximum to 1. **Fit both on train only**

Demo: Fit on Train, Apply to Test

Train cycle times stay at 28, 32 and 36 seconds

1. Can a transformed test value exceed 1? 2. Does refitting change the training rows?

Test value 40 s include test when fitting

Split	Raw seconds	Transformed
train	28	0.000
train	32	0.500
train	36	1.000
test	40	1.500

fit rows: train only · min 28 · max 36 · train 36 becomes 1.000

Answer: Reusing a Scaler

At test = 40, the saved training range produces 1.5

Fitting rows	min / max	train 36 →	test 40 →
Train only	28 / 36	1.000	1.500
Train + test	28 / 40	0.667	1.000

New observations can lie outside the fitted training range

Caveat: Min-Max and Outliers

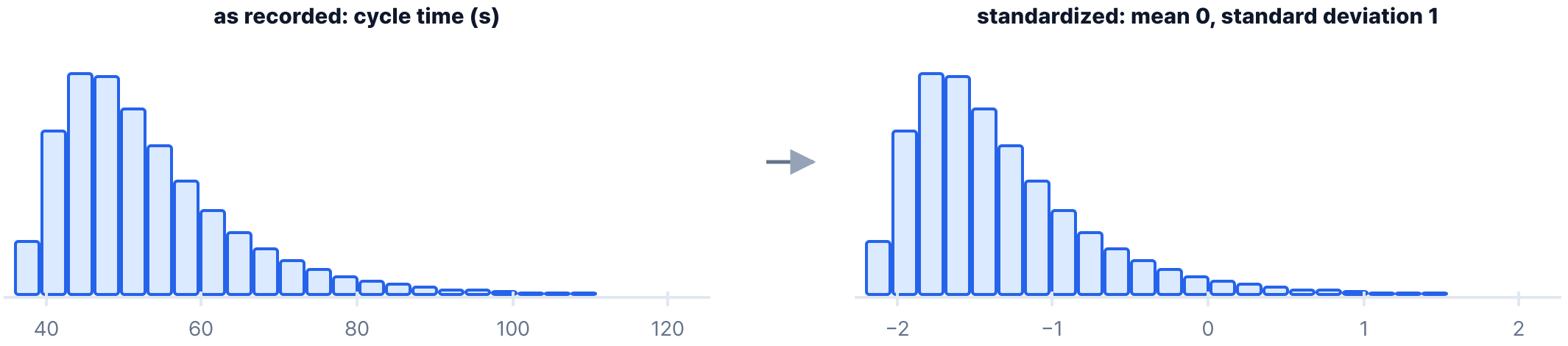
One outlier is enough to squash every other value



The maximum defines the range, so the maximum defines everything

Z-Score Standardization

Each training column gets mean 0 and standard deviation 1



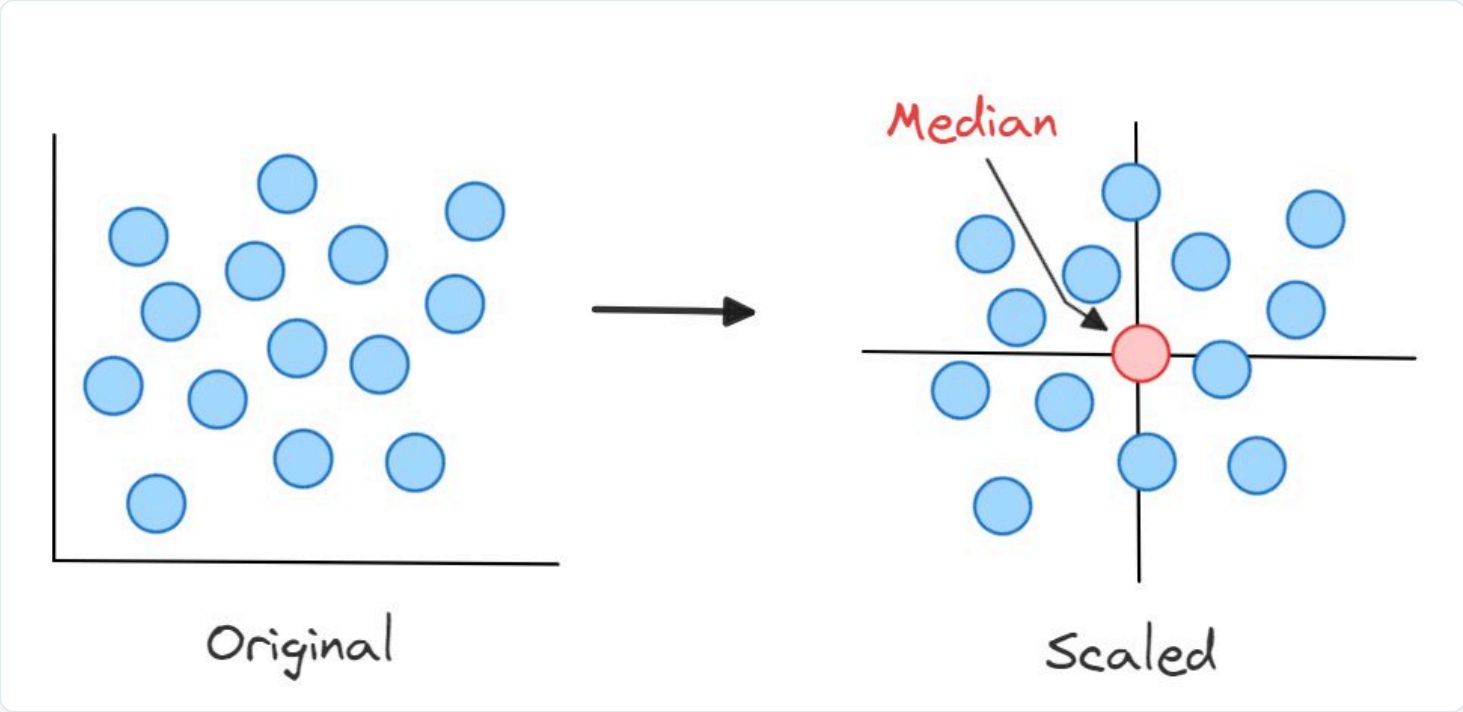
the bars are identical; only the numbers under them changed

$$z = (x - \mu) / \sigma$$ the shape is untouched; only the axis is relabeled

A right tail stays a right tail. **Standardizing does not make a column normal**

Robust Scaling

The median and the IQR barely move when one row goes wrong



$$x' = (x - \text{median}) / \text{IQR}$$ IQR = interquartile range

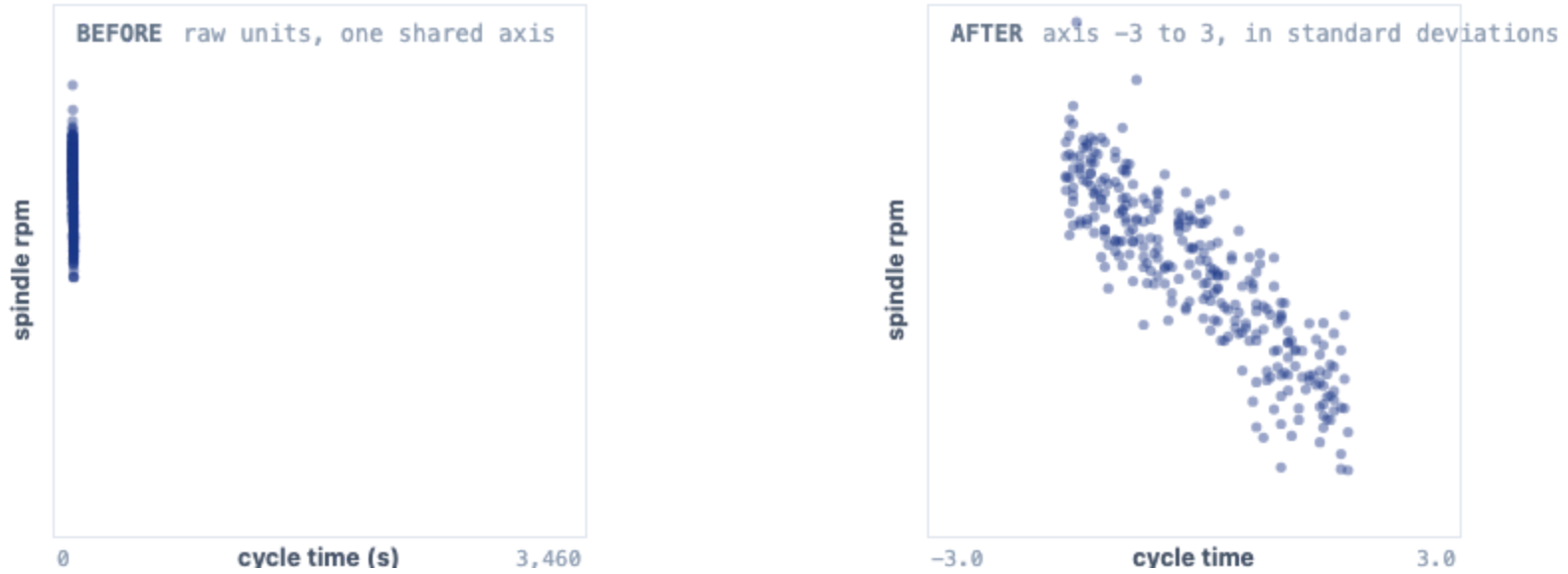
The median moves to zero and **the outliers stay far out. That is intended**

Demo: Feature Scalers

300 cycles from one computer numerical control (CNC) machine, both columns on one axis

1. Compare scalers with a mistyped rpm. 2. Did scaling repair the error or change the reference statistics?

Scaler StandardScaler ☐ one mistyped rpm (2,451 → 24,510)



raw: rpm spans **166×** the cycle-time range · after StandardScaler, the rows fill **89%** of the axis

StandardScaler is the default. RobustScaler when outliers are real. MinMaxScaler only when the bounds are known

both columns now on one scale

Answer: Scaling and Outliers

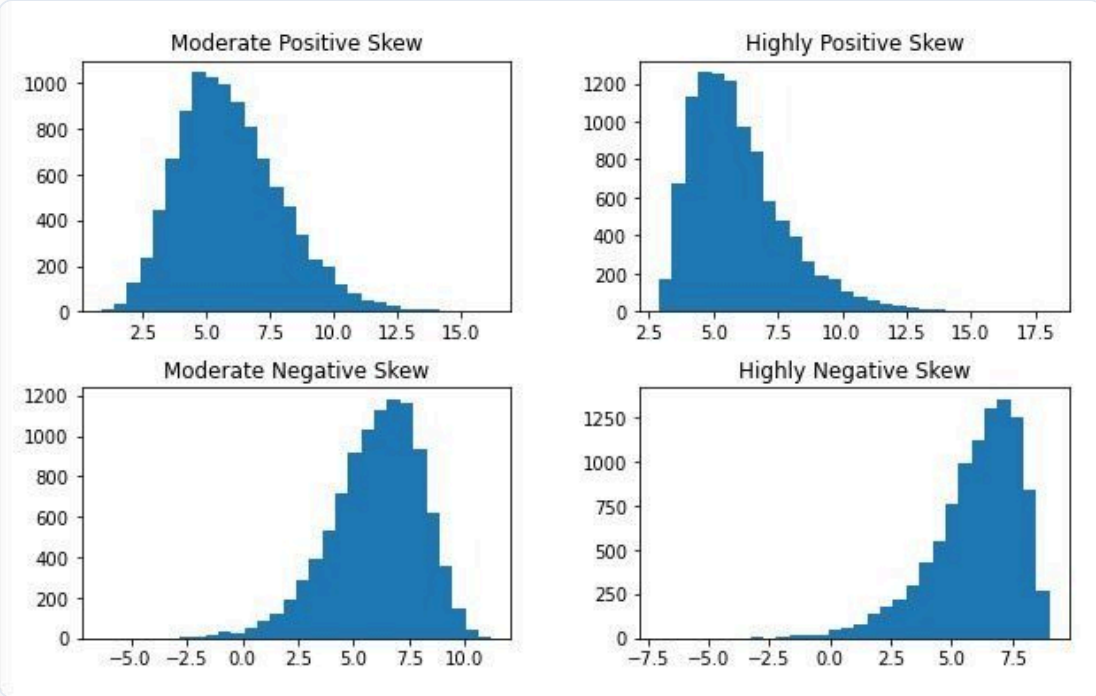
Scaling changes the reference statistics, not the recorded event

Scaler	Reference statistics	Effect of an extreme value
MinMax	min and max	clean values compress into a narrow range
Standard	mean and SD	center and spread can move
Robust	median and IQR	resists some extremes; keeps the outlier

Scaling changes representation; it does not validate or remove a value

Skewness After Scaling

Skewness survives every linear scaler

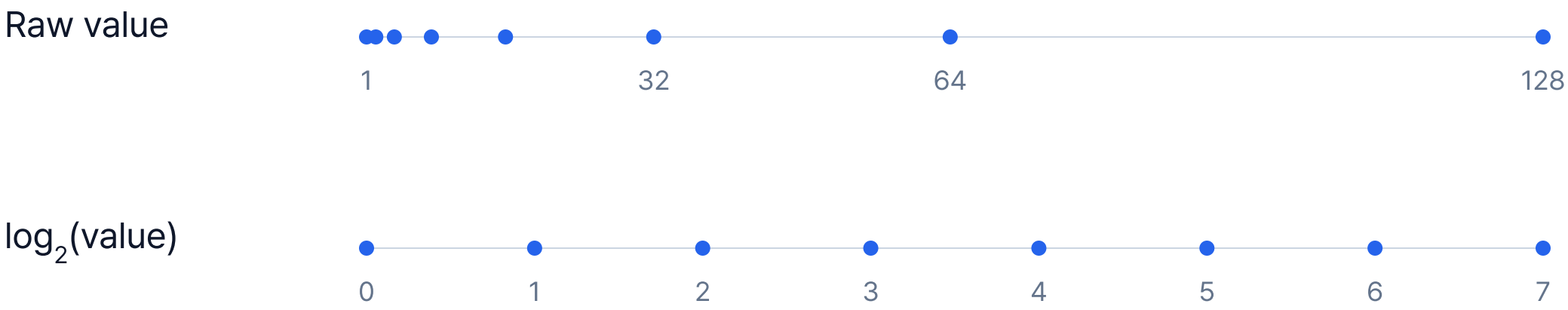


$$\gamma_1 = E\left[\left((X - \mu)/\sigma\right)^3\right]$$ positive means the long tail runs right

Positive skew runs right, negative skew runs left. **Scaling changes neither**

Log Transform

Log turns ratios into equal steps



$$\log_2(2x) - \log_2(x) = 1 \quad x > 0$$

Each doubling becomes the **same gap**. Normality is not guaranteed

Choosing a Non-Linear Transform

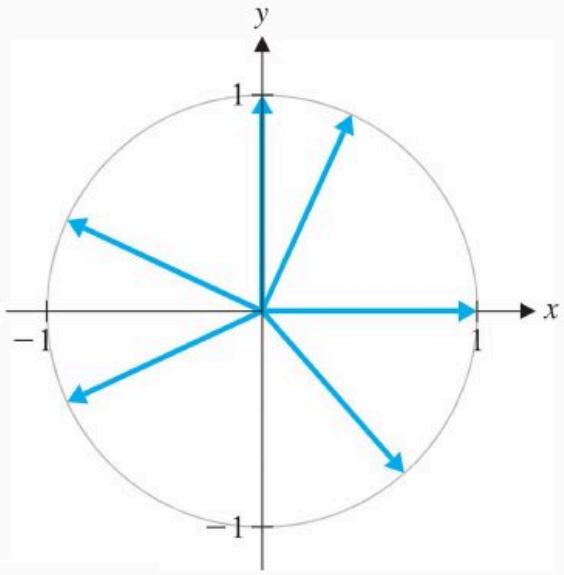
The domain of the column narrows the candidates; validation picks among them

TRANSFORM	FORMULA	DOMAIN AND USE
Log	`log(x)`	$x > 0$; multiplicative differences
Log1p	`log(1 + x)`	$x > -1$; often used for nonnegative counts
Square / cube root	$\sqrt{x}$ / $\sqrt[3]{x}$	Square root: $x \geq 0$; cube root: any real x

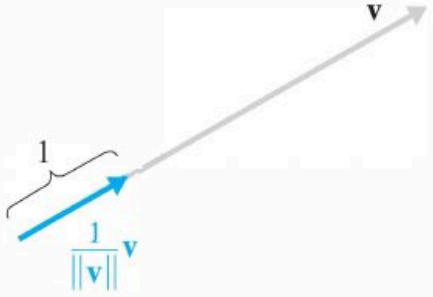
Check the input domain; evaluate whether the transformed representation helps

Unit Vector Normalization

Scale each row, not each column, to length 1



Unit vectors in $\mathbb{R}^2$

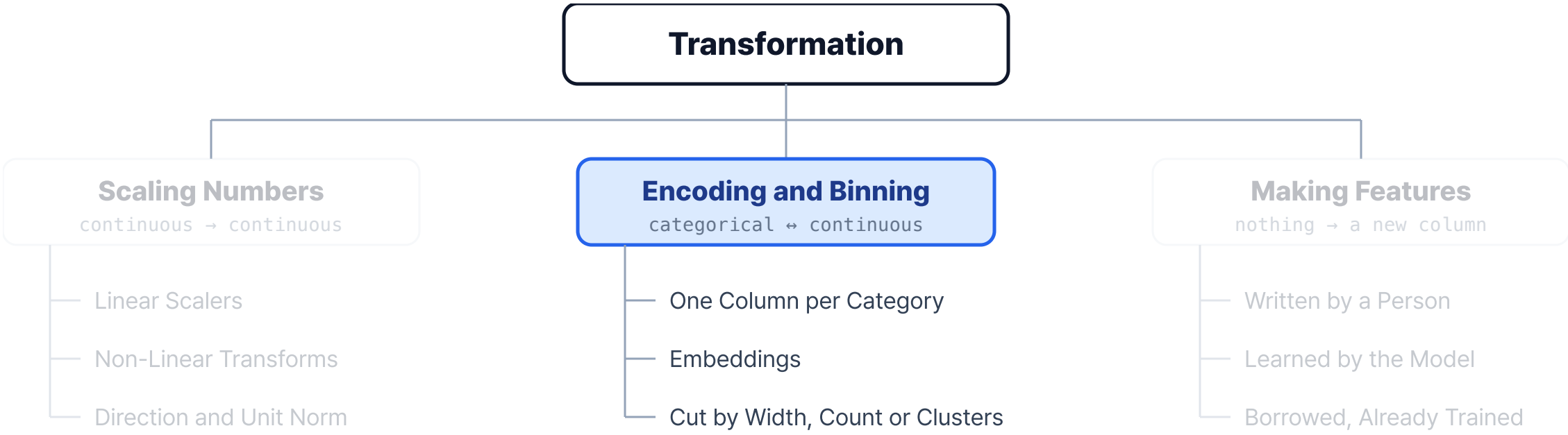


Normalizing a vector

$x' = x / \|x\|_2$ then $x'^T y'$ is exactly the cosine similarity

Every row lands on the unit circle; **the dot product is now cosine similarity**

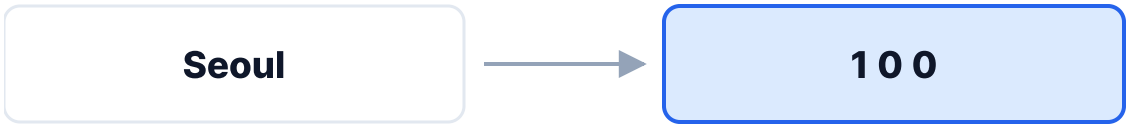
Encoding and Binning



Encoding vs Binning

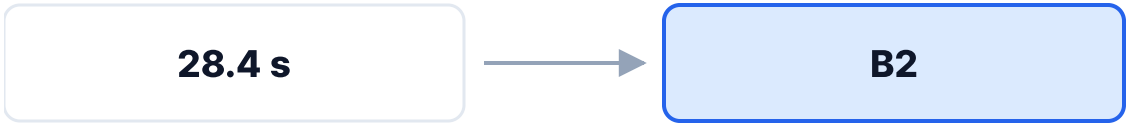
One turns categories into numbers, the other turns numbers into categories

ENCODING a model cannot multiply a name



one-hot · label · count · binary · embedding

BINNING a continuous column rarely repeats a value

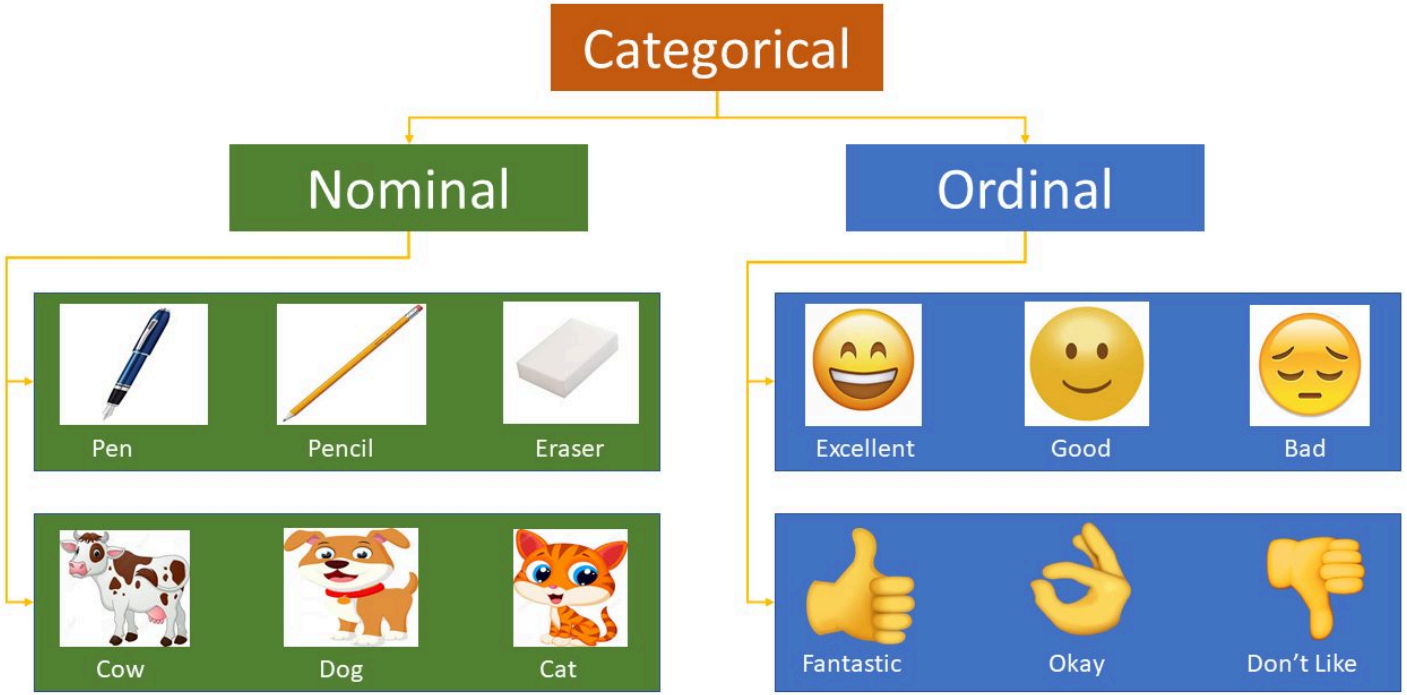


equal width · equal frequency · clustering

What the model can use decides which way you cross

Nominal or Ordinal

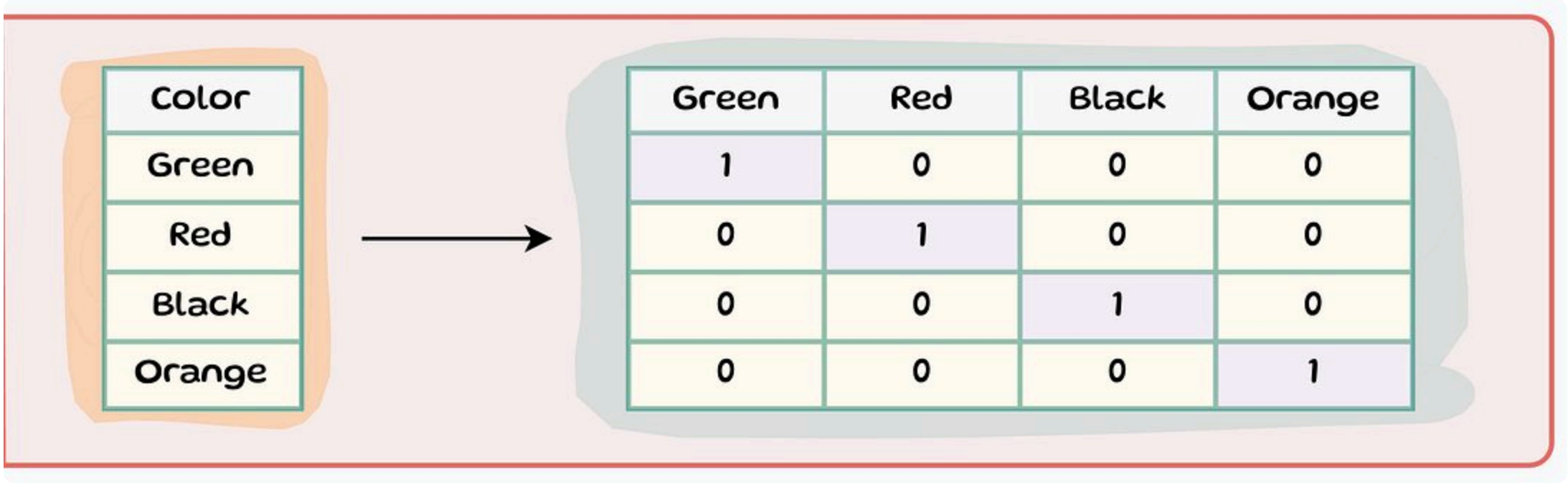
The code you pick decides what the model may compare



Integer codes impose order; **one-hot does not encode the known order**

One-Hot Encoding

One indicator column per category, so no order is invented



Color
Green
Red
Black
Orange

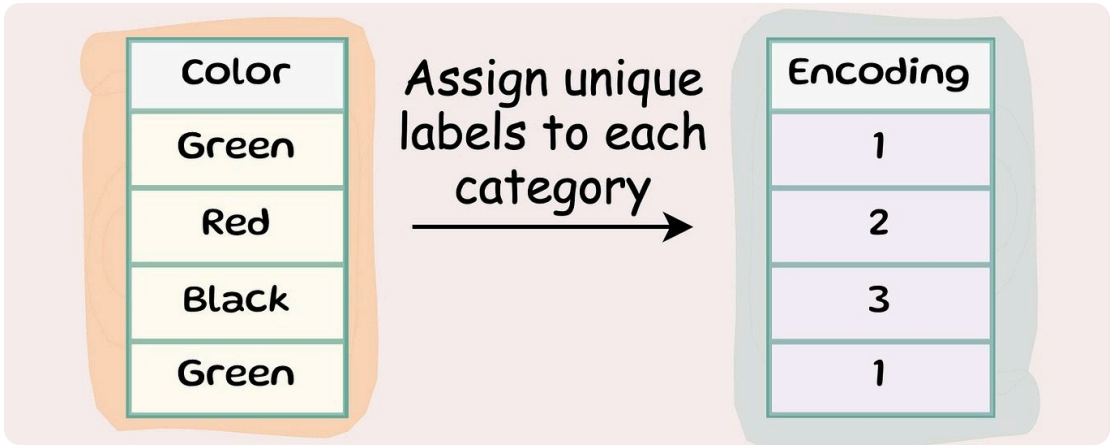
Green	Red	Black	Orange
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

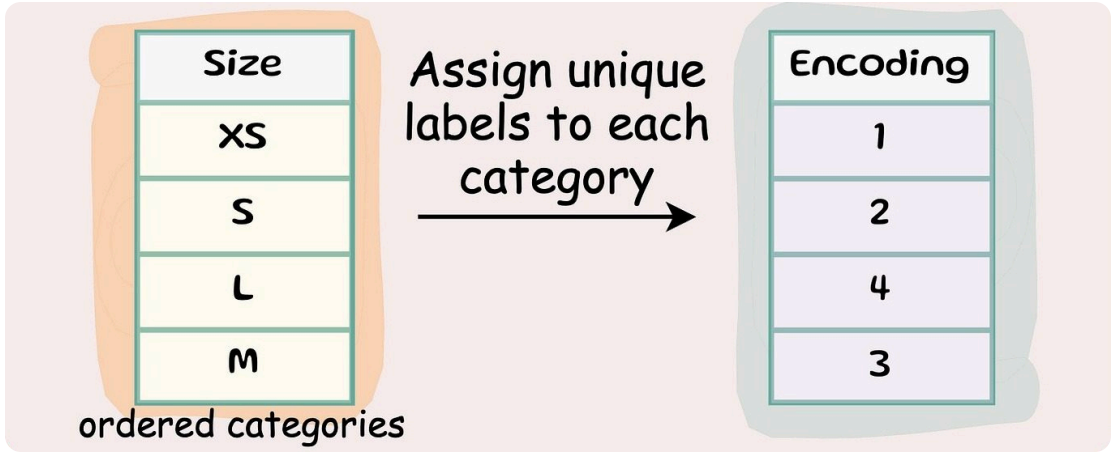
$$x \to \left(\mathbb{1}[x = c_1], \ldots, \mathbb{1}[x = c_k] \right)$$ k categories become k columns whose sum is always 1

With an intercept, **k one-hot columns are linearly dependent**. Dropping one removes this dependency

Label and Ordinal Encoding

Integer codes fit ordered categories and invent an order for unordered ones





The same trick, right once and wrong once

Choosing an Encoding

Order, the number of categories and the model decide the encoding

DATA MEANING	STARTING CHOICE	CHECK
Unordered categories	One-hot	Width and unknown values
Known ordered levels	Ordinal codes	The supplied order matches the meaning
Text or many related categories	Learned embedding	Vocabulary, model version and validation

Start with the category meaning, then check width and held-out behavior

Demo: Categorical Encoders

Six training rows; reuse their category vocabulary for test

1. Compare one-hot with integer codes. 2. Apply the fitted vocabulary to an unseen test color.

Encoder One-hot

Width estimate: total categories 4

add unseen test color: Blue

BEFORE

COLOR
Green
Red
Black
Orange
Green
Red

AFTER · fitted one-hot · unknown → zeros

BLACK	GREEN	ORANGE	RED
0	1	0	0
0	0	0	1
1	0	0	0
0	0	1	0
0	1	0	0
0	0	0	1

with 4 distinct categories: one-hot 4 columns · binary 2 · label / count 1

no order invented, 4 columns for 4 categories

Six training rows stay fixed; Blue is test only. The slider estimates width

Answer: An Unseen Category

A fitted encoder needs an explicit policy for a new value

Encoder policy	Blue at test time	Consequence
One-hot, ignore unknown	[0, 0, 0, 0]	keeps four fitted columns
Integer codes, unknown = -1	-1	unknown remains distinct
Add a test-only Blue column	five columns	changes the fitted representation

Keep the fitted vocabulary; define how unknown values are represented

Vector Embedding

A learned model maps each category to a short real vector

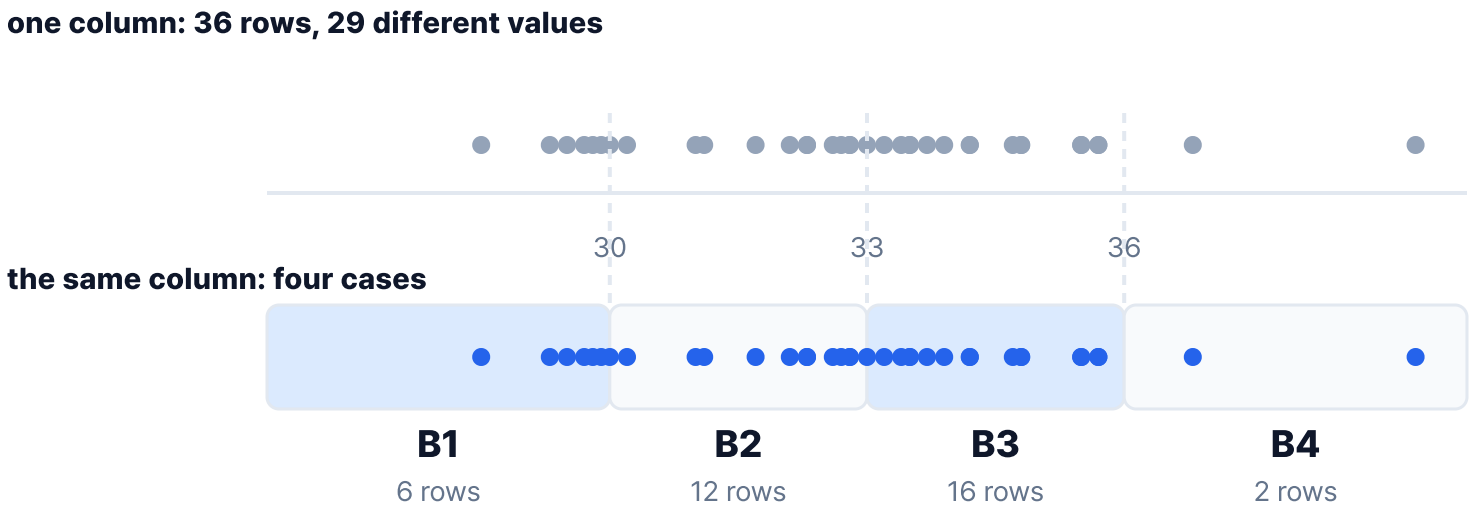


d(Seoul, Busan) = 0.35 · d(Seoul, Tokyo) = 0.83

Distances now differ between pairs. **In one-hot, every pair of categories is equally far apart**

Why Discretize

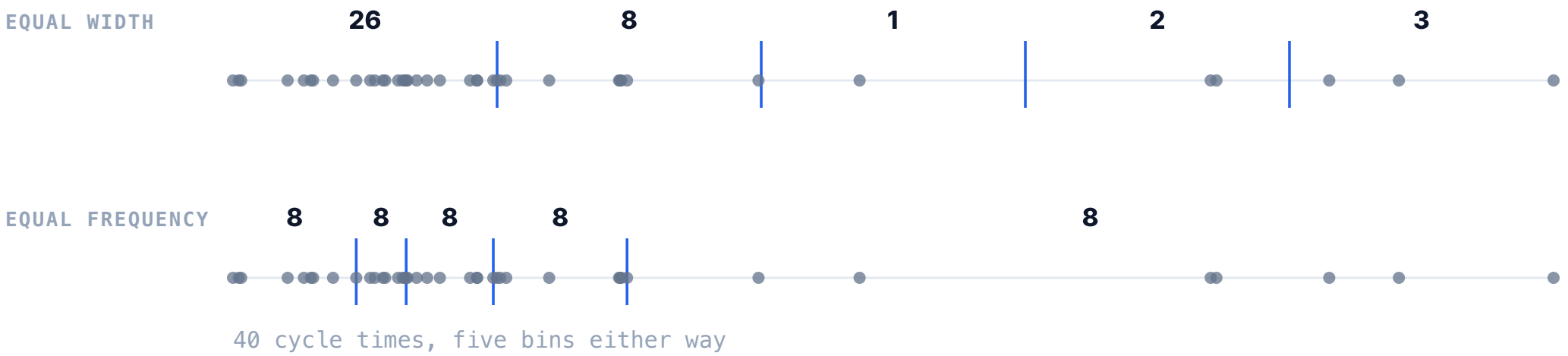
Continuous values give almost every row its own case



Four bins replace 29 values. **Where the cuts go is the whole question**

Binning by Histogram

Equal width keeps the scale, equal frequency keeps the counts



`width = (max − min) / k` vs `count = n / k` the same k, two different things held constant

Equal width: **26, 8, 1, 2, 3 rows**. Equal frequency: 8 in every bin

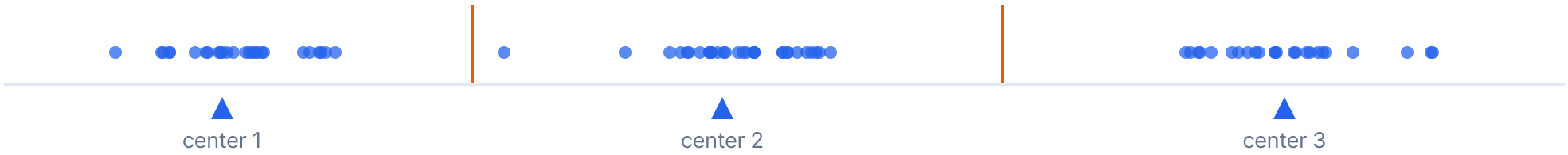
Binning by Clustering

Let the clumps in the data place the cuts

equal width: five equal bins wherever the data is

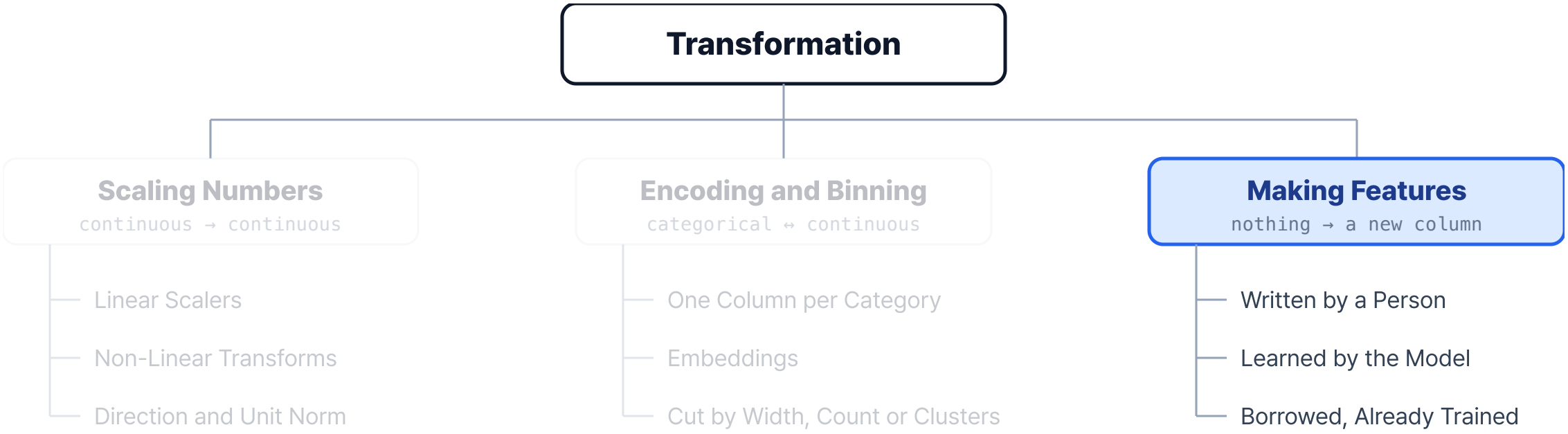


by clustering: the cuts land between the clumps



Equal width cuts through a clump; **k-means cuts in the gaps**

Making Features



Hand-Crafted Features

No column holds it, so you compute one: TF-IDF (term frequency × inverse document frequency)

TERM IN ONE MAINTENANCE LOG	TF	DF OF 1,000 LOGS	LOG(N/DF)	WEIGHT
the	12	1,000	0.00	0.0
pump	4	300	1.20	4.8
cavitation	2	12	4.42	8.8

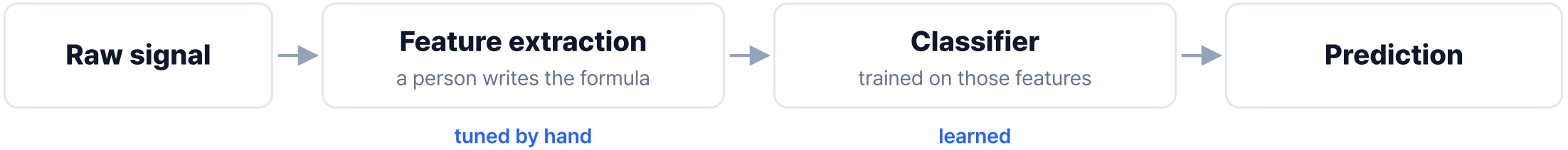
$$w = tf \cdot \log(N / df)$$ N = 1,000 logs · df = how many of them contain the term

The most frequent word scores zero. A person decided that is the right behavior

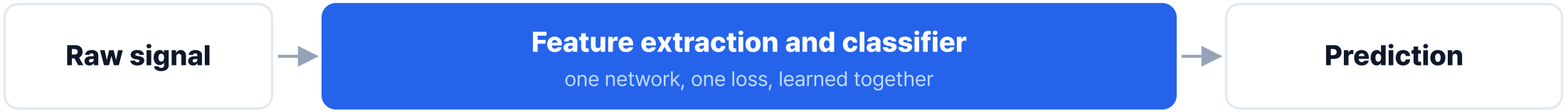
Learned Features

The network learns the features and the classifier together

CLASSIC MACHINE LEARNING



DEEP LEARNING



Fit your own encoder or reuse a versioned pretrained encoder

Case study: YouTube-8M

The dataset ships features, not videos



Feature extraction is also compression, **and it is irreversible**

Reduction

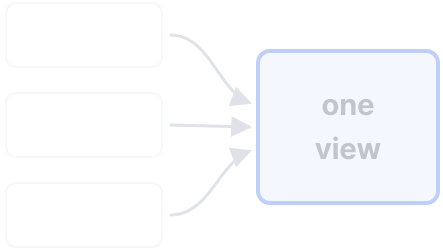
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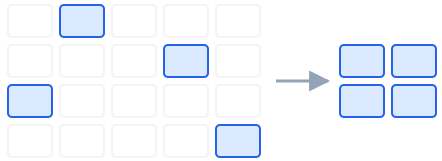
Transformation



Values a model can read

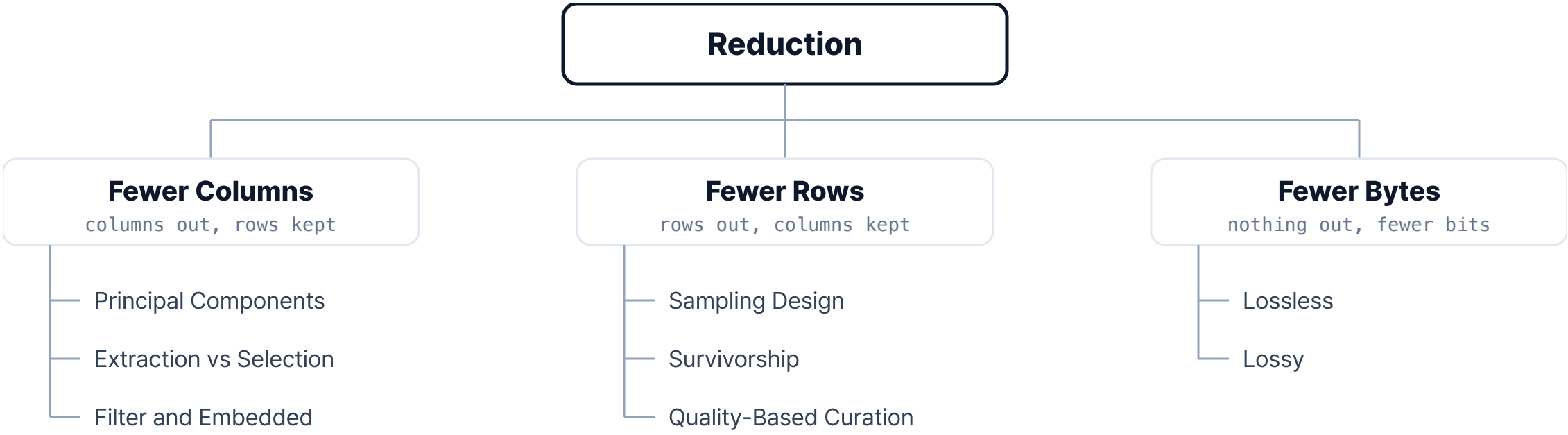
04

Reduction



Less data, same signal

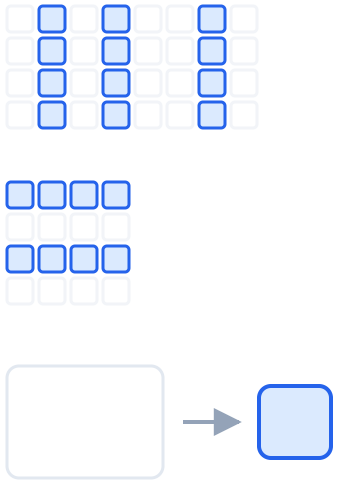
Three Branches of Reduction



Each branch removes a different thing. **Only lossless compression removes nothing**

Why Reduce at All

Columns, rows and bytes each cost something different



Fewer columns

Distances stop separating and the model fits noise

Fewer rows

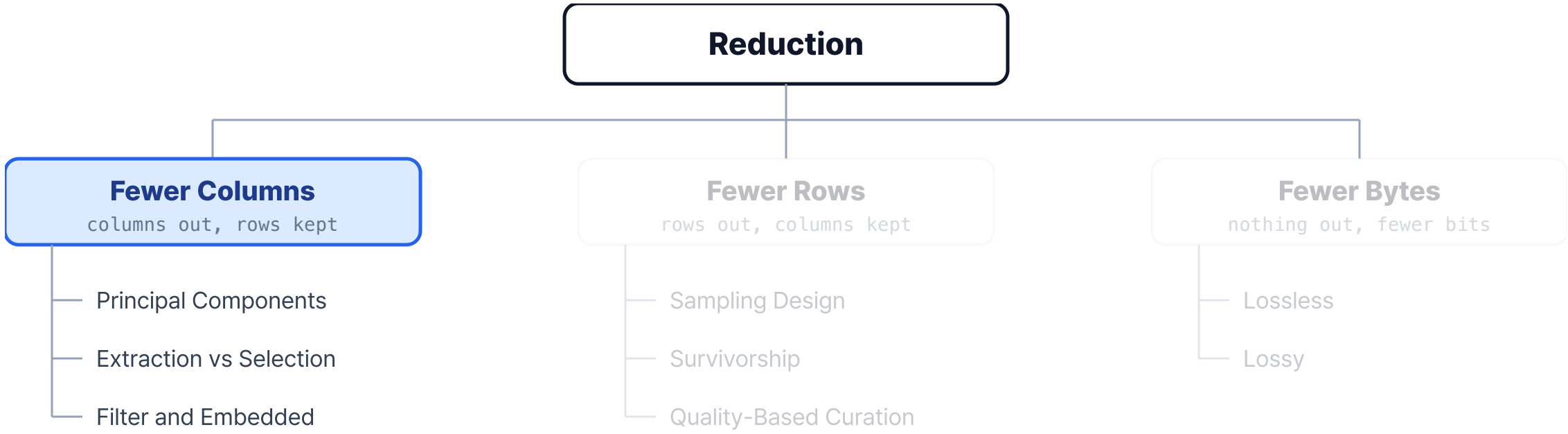
You pay to store, move and label every one of them

Fewer bytes

Storage and transfer are a bill that arrives every month

Fewer columns or rows change **what the model learns**; lossless compression changes only the bill

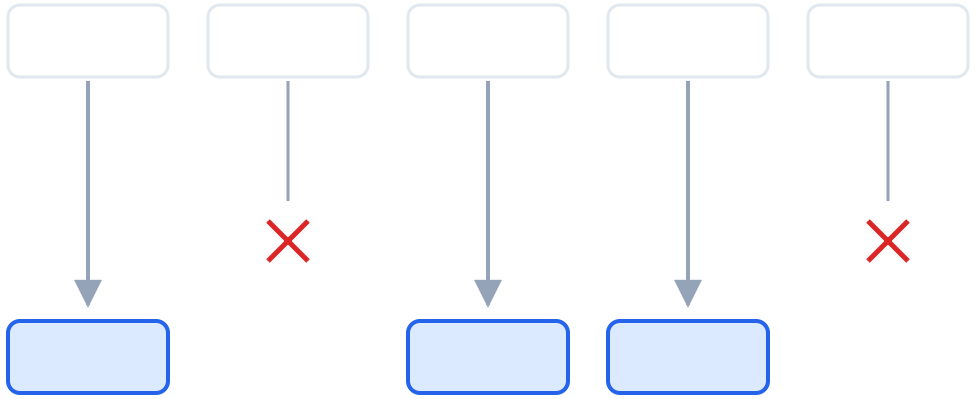
Fewer Columns



Extraction or Selection

Keep some original columns, or mix all of them into new ones

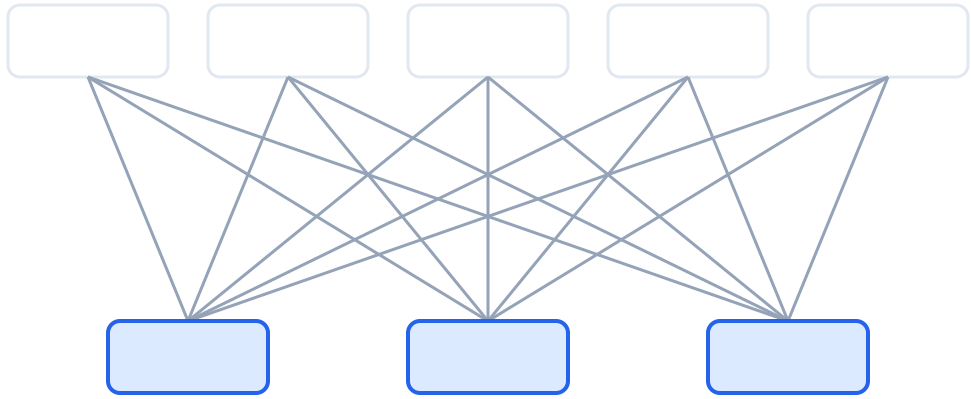
SELECTION keep some of the original columns



5 columns in, 3 of the same columns out

filter · wrapper · embedded

EXTRACTION make new columns from all of them



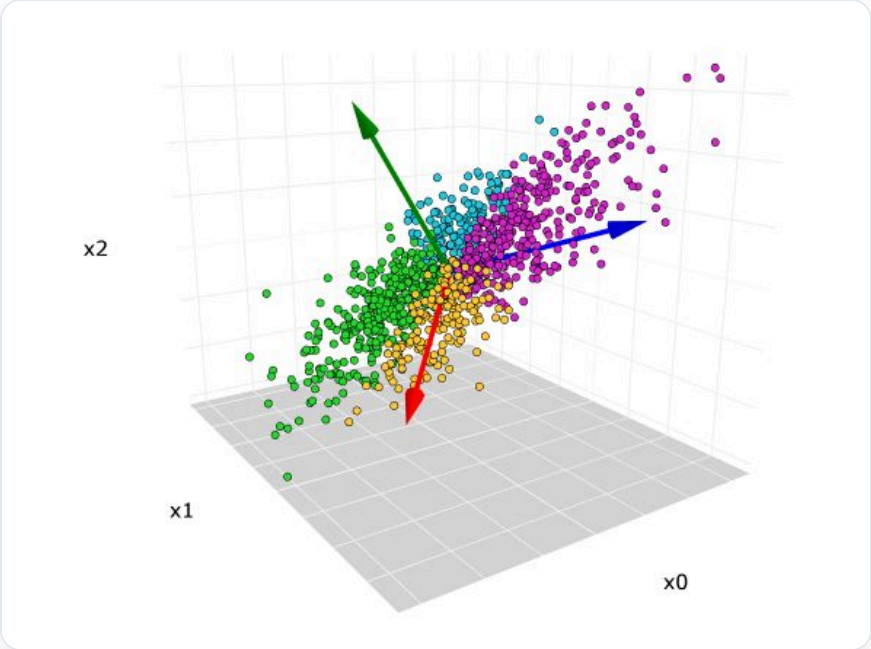
5 columns in, 3 new columns out

principal components (PCA) · random projection

Selection keeps names you can read. **Extraction keeps information you cannot name**

Recap: PCA as a Fitted Transform

Reuse the training center and axes when new observations arrive



$$z_{new} = (x_{new} - \mu_{train})\, V_{train,k}$$

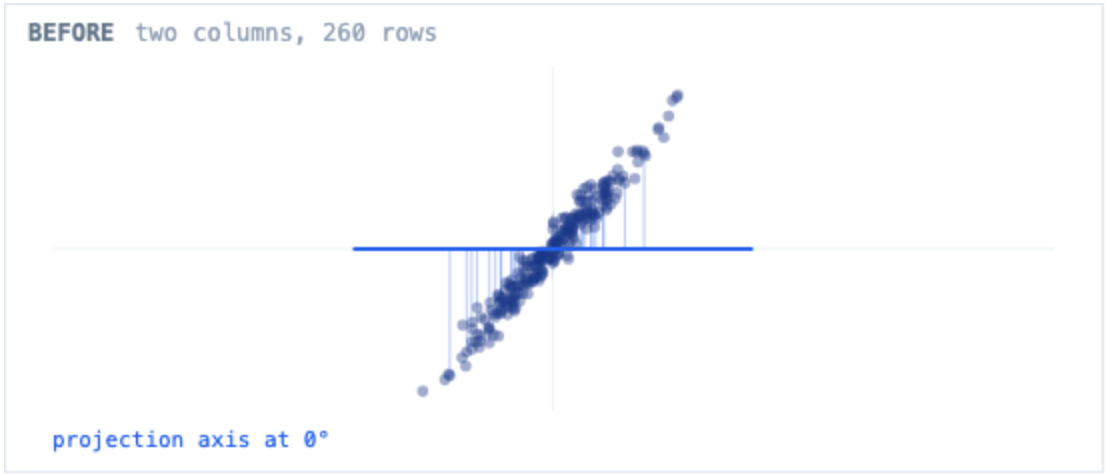
Projection keeps selected directions and discards the rest

Demo: PCA Projection Angle

Turn the axis and watch how much of the variation survives

1. Find a direction that keeps more variance. 2. Is retained variance enough to choose model features?

Projection angle 0° Use PCA





the wider these spread, the more of the original difference survives

angle **0°** keeps **43%** of the variation · PCA picks **49°** → **99%**

Center first; scale if units should count equally. Choose the component count using validation as well as retained variance

56 percentage points less variance than PCA

Answer: What PCA Keeps

Large retained variance does not guarantee useful predictive features

Question	Check
What survives?	Variation along the chosen training axes
What disappears?	Variation in discarded directions
Does the model still work?	Evaluate with held-out observations

Validate the retained representation, not only its variance

Three Families of Feature Selection

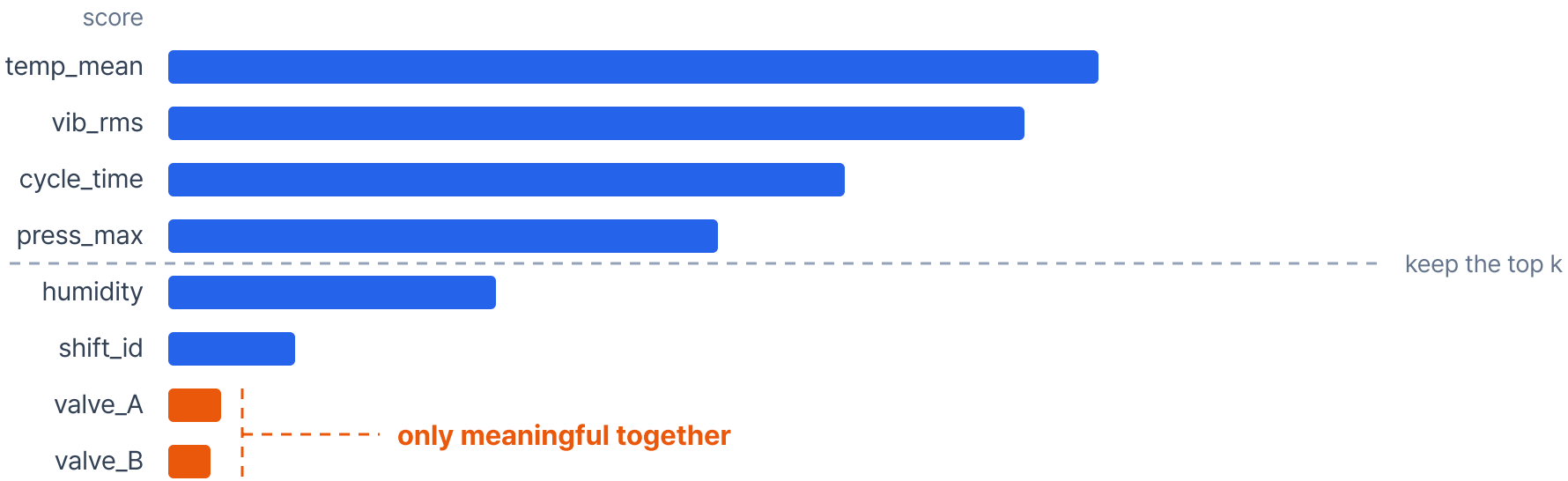
They differ in how much of the model the selection is allowed to see

FAMILY	WHAT IT LOOKS AT	MODEL FITS	BLIND SPOT
Filter	the columns alone, no model	none	a pair that only works together
Wrapper	the model’s score on a subset	many	the search itself overfits
Embedded	the model’s own coefficients	one	the answer is tied to that model

With many columns, **filter first, then embedded**. Wrapper only when the budget allows

Filter Methods

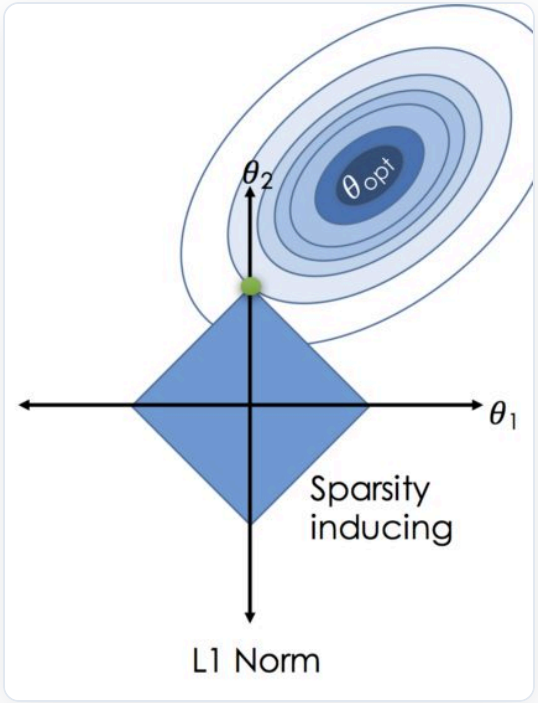
Score each column without the model: variance, correlation or mutual information



valve_A and valve_B score low alone, **yet together they predict the target**

Embedded Methods

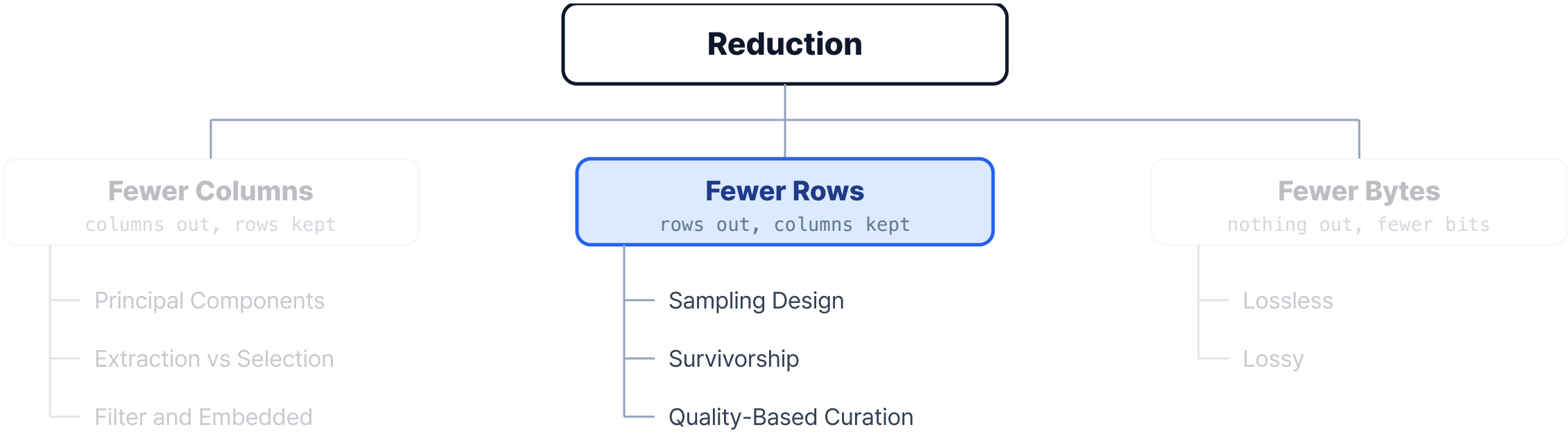
LASSO (least absolute shrinkage and selection operator) selects columns while it fits



$\min_b \|y - Xb\|^2 + \lambda\|b\|_1$ λ sets how many coefficients reach zero

The L1 constraint has corners, and a corner means a coefficient is exactly zero

Fewer Rows


Reduction
Fewer Columns
columns out, rows kept
Principal Components
Extraction vs Selection
Filter and Embedded
Fewer Rows
rows out, columns kept
Sampling Design
Survivorship
Quality-Based Curation
Fewer Bytes
nothing out, fewer bits
Lossless
Lossy

Sampling vs Curation

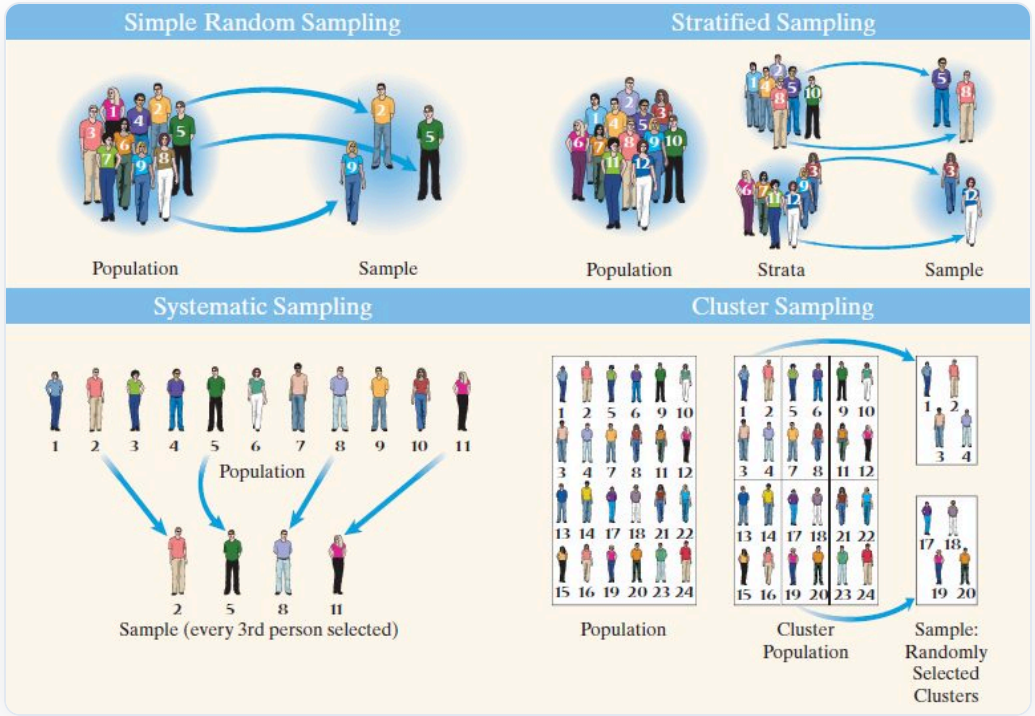
Both drop rows: sampling follows a design, curation follows quality

	SAMPLING	CURATION
Goal	A smaller set with known proportions	A better set: quality, consistency, coverage
Techniques	Random, systematic, stratified, cluster	Filtering for quality and coverage
Used for	Less computation, class imbalance	Training sets for large models, labeling budgets
Proportions	Kept, or changed on purpose (imbalance)	Changed by what the filters remove

Sampling decides **which rows to keep**; curation decides **how good they must be**

Sampling Design

Simple random, systematic, stratified and cluster sampling

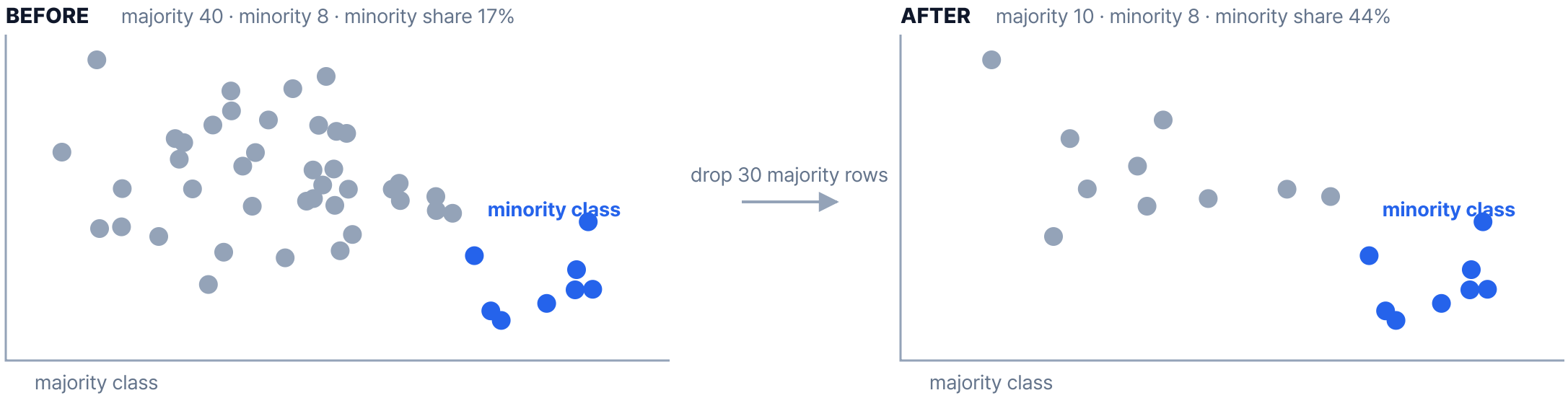


$E[\bar{x}] = \mu$ under simple random sampling, the sample mean is unbiased

Each one fails on a different structure in the population

Sampling to Fix Imbalance

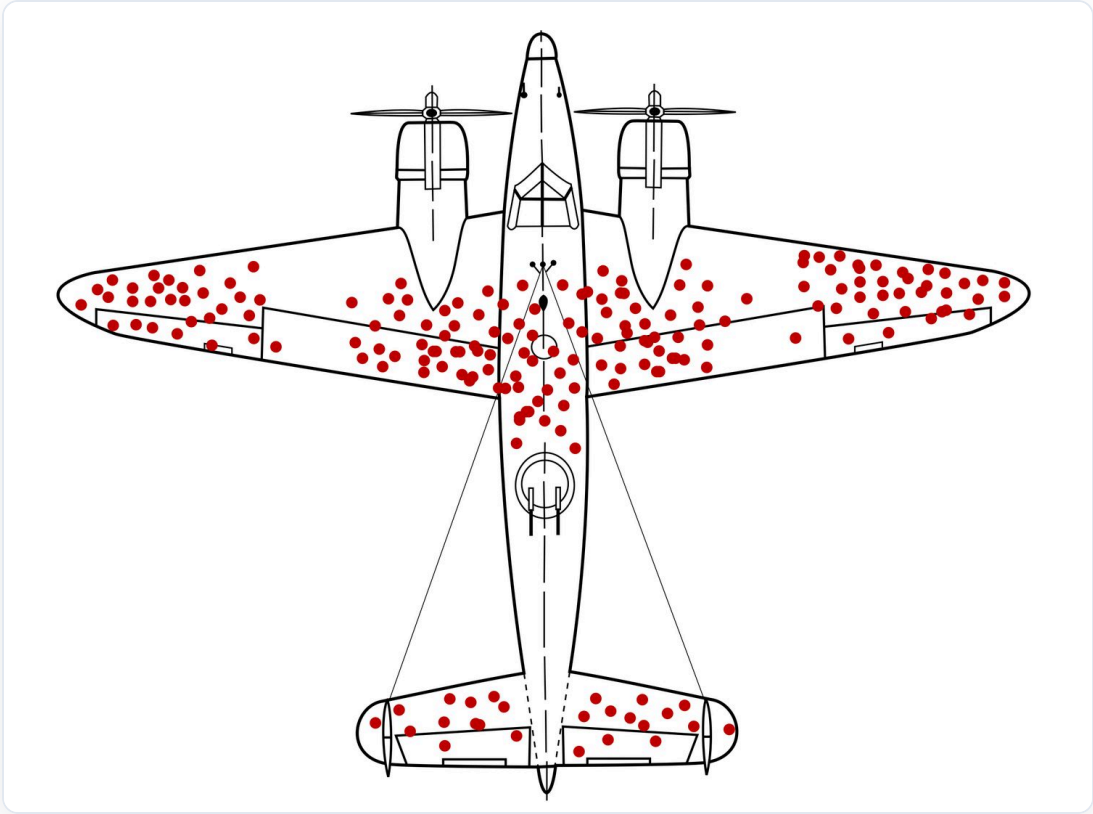
Dropping majority rows raises the minority share from 17% to 44%



Resampling changes class prevalence. **Evaluate and calibrate on the deployment distribution**

Caveat: Survivorship Bias

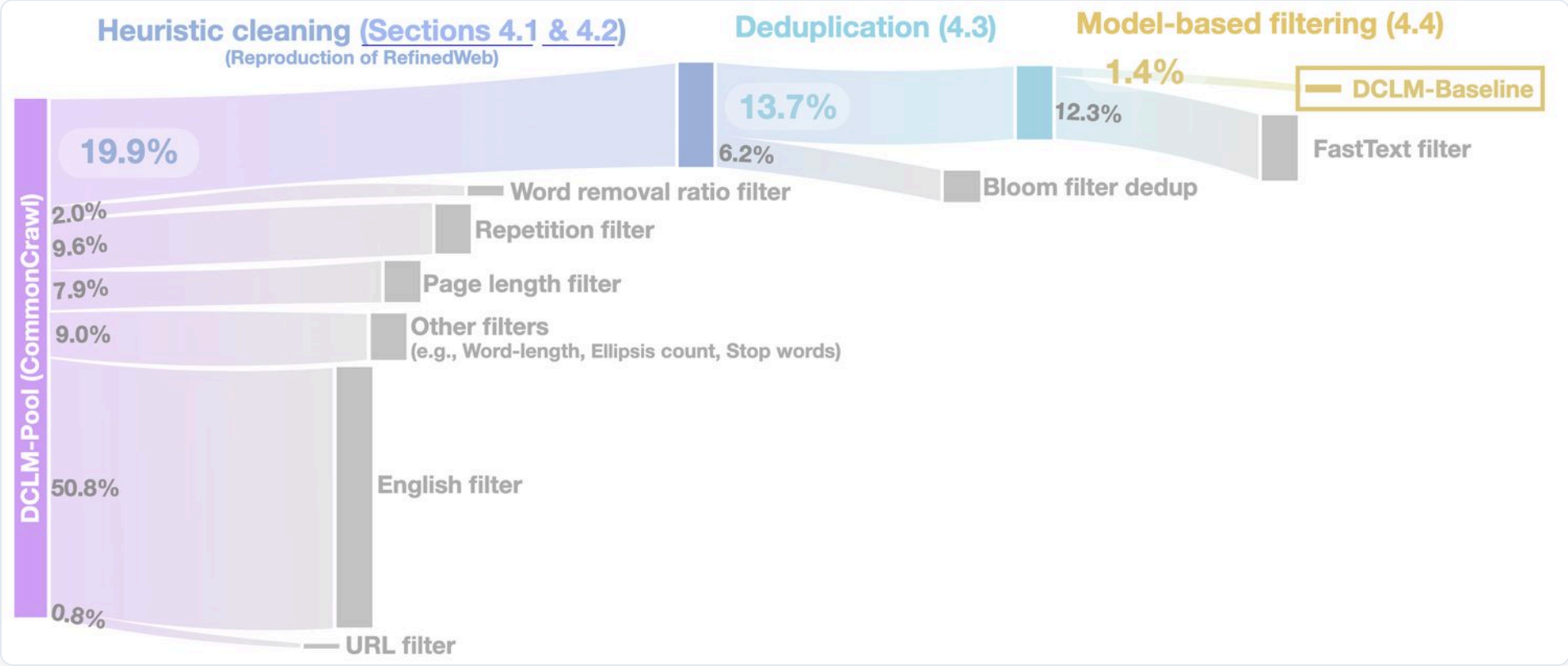
The rows you kept are not the rows you lost



This shows the bombers that came back. **Reinforce where there are no dots**

Data Curation

Curation filters for quality instead of sampling at random



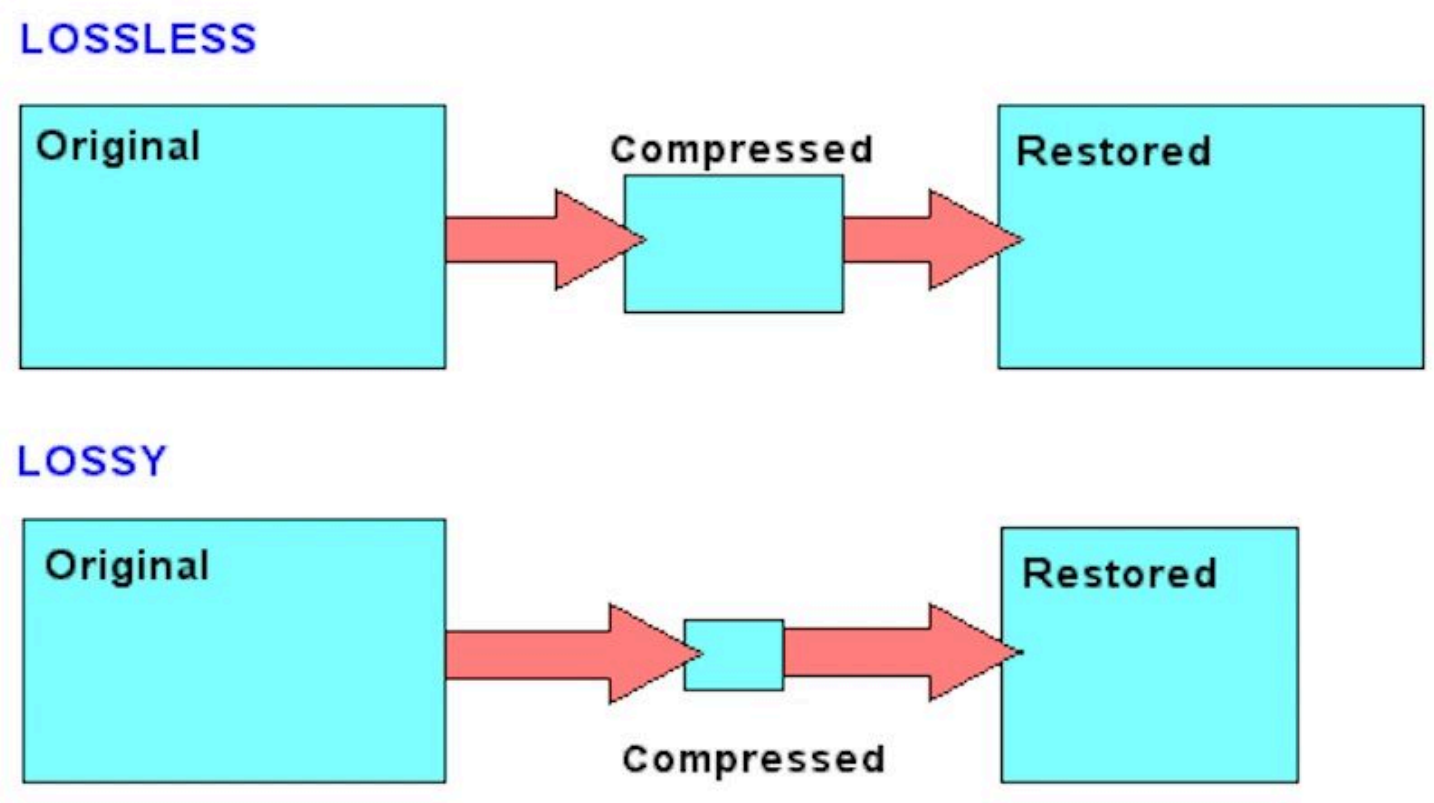
This pipeline keeps **1.4% of the crawl** as the training set

Fewer Bytes


Reduction
Fewer Columns
columns out, rows kept
Principal Components
Extraction vs Selection
Filter and Embedded
Fewer Rows
rows out, columns kept
Sampling Design
Survivorship
Quality-Based Curation
Fewer Bytes
nothing out, fewer bits
Lossless
Lossy

Lossless or Lossy

Compression restores the original bit for bit, or only an approximation



The lossy box is **smaller**, and **its changes stay**

Lossless Master Copy

Keep one lossless copy, because a lossy decision cannot be revisited

	LOSSLESS	LOSSY
Restores	The original, bit for bit	An approximation
Typical data	Text, code, logs, tabular	Images, audio, video
Formats	ZIP, PNG, GZIP, Parquet with Snappy or Zstd	JPEG, MP3; lossy H.264 / WebP
Ratio	Limited by entropy	Often much higher, tunable
The catch	Sometimes not small enough	The decision cannot be revisited

Text and logs stay lossless; **media is where lossy pays**

Fitting Preprocessing Steps

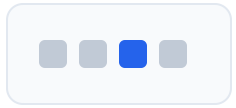
Fit on train, apply to test: every fitted step remembers something from the rows it saw

STEP	WHAT IT REMEMBERS	WHAT LEAKS IF YOU FIT ON EVERYTHING
Imputation	the column median	test rows move the median
Scaling	mean and standard deviation	the test spread enters every train row
Encoding	the list of categories	a category that exists only in test
Binning	the bin edges	edges drawn around test values
Feature selection	the columns kept	columns kept because they help on test
PCA	the components	axes rotated to fit test variance

$$z = (x - \mu_{train}) / \sigma_{train}$$ the same two numbers go to test and to every future row

Refit inside **every fold**; ship the final fit to testing and serving

Takeaway: Every transform and reduction is a decision



A scaler or an encoding decides what the model may compare

The center, the spread and any order you invent all end up inside the model



Know which reductions you can undo

Lossless compression restores; dropped rows and quantized values do not



Every fitted step carries state

Fit on train, apply to test, and ship the fitted object with the model